1 Smoothed Particle Hydrodynamics

This chapter presents the main numerical method used — the *Smoothed Particle Method*. In particular,

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1.1 Method overview

Smoothed Particle Hydrodynamics (SPH) is a general use numerical method first used in astrophysical stellar models [1]. The authors required a solution to a grid resolution problem arising with their use of the finite difference method (FDM).[1] The method has found use in a variety of fields, viz. free-surface flows [2], solid mechanics [3], geomechanics [4], and is still successfully used in astrophysics [5].

SPH is a particle (Lagrangian) method.[2] The continuum is partitioned into arbitrary *material points*, in this context referred to as *particles*, and basic physical properties such as density, velocity, entropy, and energy are assigned to them. Their mutual interaction is then prescribed,[3] and the state of each particle is advanced in time. We stress that *SPH particles* are *not* elementary particles in the microscopic sense — they are typically macroscopic fluid elements. Nonetheless, some intuition from the mechanics of point particles carries over naturally to SPH, as we shall see.

In what follows, we introduce the theoretical foundations of SPH and the specific formulation developed in this work.

1.2 Cornerstones of SPH

In this section, we outline the standard approach to SPH discretisation.

Kernel interpolation

Let $\Omega \subset \mathbb{R}^d, d = 2, 3$ be the continuous medium and $\Phi : \mathbb{R}_0^+ \times \Omega \to \mathbb{R}$ an arbitrary scalar valued field, e.g., density. Suppose further that $\forall t \in \mathbb{R}_0^+$ the mapping $\mathbf{x} \mapsto \Phi(t, \mathbf{x})$ is in $L^1_{\text{loc}}(\Omega)$, then

$$\Phi(t, \mathbf{x}) = (\Phi \star \delta)(t, \mathbf{x}) = \int_\Omega \Phi(t, \mathbf{x}')\delta(\mathbf{x} - \mathbf{x}')\, \mathrm{d}\mathbf{x}' \,, \tag{1.1}$$

[1] They argued that the treatment of highly distorted continuum would require an extremely fine grid, hence substantial computational cost p. 1 [1]

holds for almost all $\mathbf{x} \in \Omega$. A common assumption in the SPH literature is to assume $\mathbf{x} \mapsto \Phi(t, \mathbf{x}) \in C(\Omega)$, and so the equality Equation (1.1)$^{\to \text{p.2}}$ holds pointwise in Ω. To make the presentation simpler, throughout we always assume sufficient regularity of all quantities involved and do not pursue measure-theoretic rigor.

The main point of departure of the SPH method is to replace the Dirac delta distribution in Equation (1.1)$^{\to \text{p.2}}$ by an *interpolation kernel* $w_h : \Omega \to \mathbb{R}$ p. 305 [8], i.e.,

$$[\Phi]_c(t, \mathbf{x}) := (\Phi \star w_h)(t, \mathbf{x}) = \int_\Omega \Phi(t, \mathbf{x}') w_h(\mathbf{x} - \mathbf{x}') \, \mathrm{d}\mathbf{x}' \,. \tag{1.2}$$

The mapping $\Phi \mapsto [\Phi]_c$ is called *continuous interpolation*. Certainly, the interpolation kernel w_h cannot be arbitrary. We aim to approximate the function Φ by its continuous interpolation $[\Phi]_c$, that is

$$\Phi(t, \mathbf{x}) \approx [\Phi]_c(t, \mathbf{x}) = \int_\Omega \Phi(t, \mathbf{x}') w_h(\mathbf{x} - \mathbf{x}') \, \mathrm{d}\mathbf{x}' \,.$$

To obtain a good approximation, the kernel must satisfy p. 60 [9] the conditions written out in Table 1.3$^{\to \text{p.2}}$.

	Property	Condition
(K1)	positivity	$w_h \geq 0$
(K2)	monotonicity	w_h is a decreasing function
(K3)	symmetry	w_h is radial, i.e., $w_h(\mathbf{x} - \mathbf{x}') = w_h(\lvert\mathbf{x} - \mathbf{x}'\rvert)$
(K4)	normalization	$\int_\Omega w_h(\mathbf{r}) \, \mathrm{d}\mathbf{r} = 1$
(K5)	compact support/decay	$\operatorname{supp} w_h$ is compact or $\lim_{\lvert\mathbf{r}\rvert \to \infty} w(\mathbf{r}) = 0$ sufficiently fast
(K6)	smoothness	w_h be *sufficiently smooth*
(K7)	concentration	$w_h \overset{*}{\rightharpoonup} \delta$ as $h \to 0^+$

Table 1.3: Properties of the interpolation kernel w_h.

For compact kernels, h is related to the size of the support radius as $\operatorname{diam} \operatorname{supp} = 2\kappa h$, where κ is kernel-dependent. If w_h is not compact, e.g., for a Gaussian kernel p. 8 [10]

$$w_h(\mathbf{r}) = \frac{1}{(h\sqrt{\pi})^d} \exp\left(-\frac{\lvert\mathbf{r}\rvert^2}{h^2}\right),$$

h is the characteristic width of the Gaussian.

Common choice for the interpolation kernel are "bell shaped" functions, such as the Gaussian, B-splines [11] and other splines. An overview of the commonly used kernels can be found in p. 62 [9].

In our application, we deployed the C^2 Wendland polynomial kernels [12] [13], i.e., a special class of piecewise polynomial positive compactly supported functions.[4] In particular, the quintic Wendland kernel has been used

$$w_h(q) = \begin{cases} \frac{7}{4\pi h^2}\left(1-\frac{q}{2}\right)^4(2q+1), & 0 \le q \le 2, \\ 0, & q > 2, \end{cases}, q = \frac{|\mathbf{r}|}{h}. \tag{1.4}$$

The graph of the Wendland kernel is plotted in Figure 1.5 → p. 3. It is evident that for the Wendland kernel Equation (1.4) → p. 3, $w_h\left(\frac{\mathbf{r}}{h}\right) = 0$ for $|\mathbf{r}| \le 2h$, meaning $4h = \operatorname{diam}\operatorname{supp} w_h$ and we call h the *smoothing length*[5]

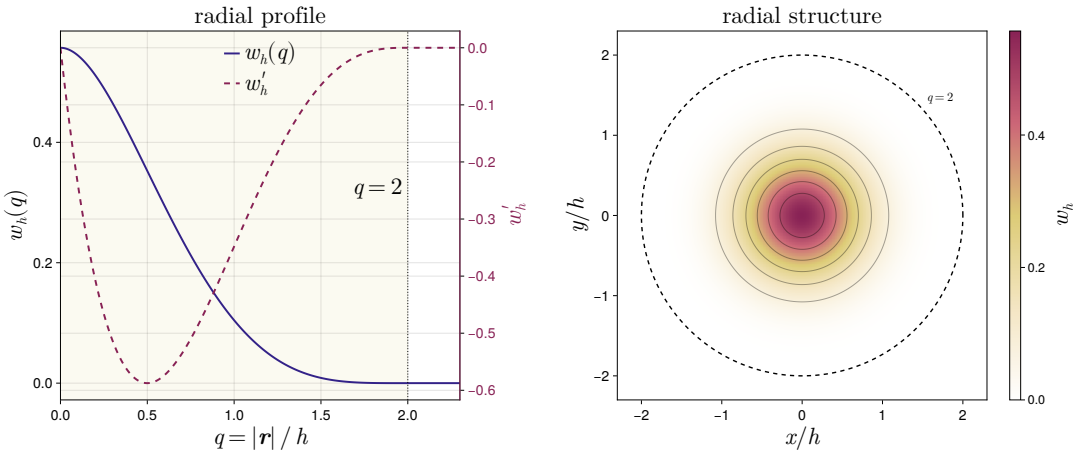

Figure 1.5: Wendland quintic kernel, $h = 1$.

Under the conditions Table 1.3 → p. 2 on the interpolation kernel w_h, it can be shown[6] p. 307 [8]

$$[\Phi]_c(t, \mathbf{x}) = \Phi(t, \mathbf{x}) + \mathcal{O}\left(h^2\right). \tag{1.6}$$

This approximation property evidently heavily influences the numerical accuracy of the method. A more involved discussion of SPH accuracy and consistency will be presented in section 9 → p. 5.

Particle approximation

Let $\Omega \subset \mathbb{R}^d, d = 2, 3$ be a continuous medium, which we partition into $N \in \mathbb{N}$ particles with coordinates $\{\mathbf{x}_\alpha\}_{\alpha=1}^N \subset \mathbb{R}^{dN}$. Denote by V_α some *estimator* of the particle volume, e.g.,

$$V_\alpha = \frac{\rho_\alpha}{m_\alpha}\,, \tag{1.7}$$

with ρ_α being the particles density and m_α its mass. The choice Equation (1.7) → p. 4 is clearly *not a unique* choice of the volume estimator; the freedom of choice will be invoked in Using this particle partition, we simplify the continuous interpolation Equation (1.2) → p. 2 by deploying a quadrature rule. Let $\mathbf{x}_\alpha$ be the position of an arbitrary particle, then we approximate

$$[\Phi]_c(t, \mathbf{x}_\alpha) = \int_\Omega \Phi(t, \mathbf{x}')w_h(|\mathbf{x}_\alpha - \mathbf{x}'|)\,\mathrm{d}\mathbf{x}' \approx \sum_{\beta=1}^{N} \Phi(t, \mathbf{x}_\beta)w_h(|\mathbf{x}_\alpha - \mathbf{x}_\beta|)V_\beta,$$

that is, we approximate the integration by a *quadrature rule* with nodes $\{\mathbf{x}_\beta\}_{\beta=1}^N$ and weights $\{V_\beta\}_{\beta=1}^N$. This procedure can be thought of as introducing a *discrete interpolation* operator $\Phi(t, \mathbf{x}_\alpha) \mapsto [\Phi]_d(t, \mathbf{x}_\alpha)$ given as

$$[\Phi]_d(t, \mathbf{x}_\alpha) \coloneqq \sum_{\beta=1}^{N} \Phi(t, \mathbf{x}_\beta)w_h(|\mathbf{x}_\alpha - \mathbf{x}_\beta|)V_\beta \equiv \sum_{\beta=1}^{N} \Phi_\beta w_{\alpha\beta}V_\beta, \tag{1.8}$$

where we have adopted the notation $\Phi_\beta \coloneqq \Phi(t, \mathbf{x}_\beta)$, $w_{\alpha\beta} \coloneqq w_h(|\mathbf{x}_\alpha - \mathbf{x}_\beta|)$. The quantity Φ_β can be naturally interpreted as the quantity Φ, i.e., density, of the particle $\mathbf{x}_\beta$. It should be clarified that $\Phi_\beta = \Phi_\beta(t)$ is still a function of time. Notice also that since the kernel w_h has a compact support, in practice the sum in Equation (1.8) → p. 4 typically does *not* consist of N summands.[7]

Having introduced the (discrete) interpolation, we deploy it to approximate the *continuous* interpolation

$$[\Phi]_c(t, \mathbf{x}_\alpha) \approx [\Phi]_d(t, \mathbf{x}_\alpha),$$

and one would immediately be interested in the quality of the approximation. It turns out that an involved discrete approximation error is a complicated *open* problem p. 3 [14] [15]. The main difficulty lies in the fact that the error analysis depends on the particle positions (that change in time). If one considers particles evenly arranged in a Cartesian grid (which is a substantial simplification), it can be shown[8] that for the Wendland quintic kernel it holds

$$[\Phi]_d(t, \mathbf{x}) = [\Phi]_c(t, \mathbf{x}) + \mathcal{O}\left(\left(\frac{\mathrm{d}r}{h}\right)^6\right), \tag{1.9}$$

with $\mathrm{d}r$ being the typical particle size, i.e., $\mathrm{d}r \propto V^{\frac{1}{d}}$, for particles of typical volume V.

Accuracy of SPH

A presentation of a numerical method would not be complete without a mention of the *error estimation* — that is, what accuracy the method achieves.

There are two sources of errors in the SPH discretisation: **(1)** continuous kernel interpolation, **(2)** discrete particle approximation , which we analysed in Equation (1.6) →p.3 and Equation (1.9) →p.5 respectively. Combining these, we obtain for a sufficiently smooth scalar field Φ, the quintic Wendland kernel of smoothing length h, and particles of size $\mathrm{d}r$ arranged on a regular Cartesian grid,

$$[\Phi]_d = \Phi + \mathcal{O}\left(h^2\right) + \mathcal{O}\left(\left(\frac{\mathrm{d}r}{h}\right)^6\right). \tag{1.10}$$

This relation shows that the SPH method is *at least first order*. We say "at least" because for particular combinations of kernel, geometry and particle arrangement, the observed order can be higher [15]. Moreover, the error decreases when both $h \to 0^+$ and $\mathrm{d}r\,/h \to 0^+$, equivalently $h/\,\mathrm{d}r \to \infty$. This reveals an interesting relationship between the smoothing length h and the particle size $\mathrm{d}r$: to improve accuracy, both must vanish[9], but *not independently* — $\mathrm{d}r$ must decay faster than h. The precise interplay of these two parameters remains an active area of analysis [15].

Let us summarize the presented properties and situate SPH within the broader landscape of numerical methods for continuum mechanics. Mesh-based Eulerian methods such as the finite volume method and, in particular, the discontinuous Galerkin method can be constructed to achieve arbitrarily high order of convergence, at least in *smooth regions of the solution*.[10] In this comparison, SPH — which is often merely first order in practice — is clearly less competitive.

However, accuracy should not be the only criterion. The mesh-free nature of SPH allows us to tackle problems that are notoriously difficult for mesh schemes: free-surface flows, large deformations, plasticity, flows in complex or evolving geometries. No meshing routines are required. The method is also remarkably simple to implement.

In summary, SPH trades numerical accuracy for flexibility and implementational simplicity. The choice of method is therefore dictated by the nature of the problem under study.

Differential operators

This section employs the presented particle approximation framework to discretise differential operators appearing in the governing equations of fluid dynamics. Even

though we have, up to this point, assumed only scalar fields, the generalization to vector and tensor fields is straightforward.

Gradient of a scalar field

The gradient of a differentiable scalar field $\Phi(t, \mathbf{x})$ at particle $\mathbf{x}_\alpha$ can be in the nature of Equation (1.8) → p. 4 approximated as

$$\begin{aligned}\nabla_\alpha \Phi(t, \mathbf{x}_\alpha) \approx [\nabla_\alpha \Phi]_d(t, \mathbf{x}_\alpha) &= \sum_{\beta=1}^{N} \nabla_\alpha (\Phi(t, \mathbf{x}_\beta) w_h(|\mathbf{x}_\alpha - \mathbf{x}_\beta|)) V_\beta = \\ &= \sum_{\beta=1}^{N} \Phi(t, \mathbf{x}_\beta) \nabla_\alpha w_h(|\mathbf{x}_\alpha - \mathbf{x}_\beta|) V_\beta = \\ &= \sum_{\beta=1}^{N} \Phi_\beta \nabla_\alpha w_{\alpha\beta} V_\beta,\end{aligned}$$

since $\Phi(t, \mathbf{x}_\beta)$ is independent of $\mathbf{x}_\alpha$,

Divergence of a vector field

The discrete divergence of a continuously differentiable vector field $\mathbf{F}(t, \mathbf{x})$ at a particle $\mathbf{x}_\alpha$ is defined as

$$\begin{aligned}\nabla_\alpha \cdot \mathbf{F}(t, \mathbf{x}_\alpha) \approx [\nabla_\alpha \cdot \mathbf{F}]_d(t, \mathbf{x}_\alpha) &= \sum_{\beta=1}^{N} \Big(\nabla_\alpha \Big(\mathbf{F}(t, \mathbf{x}_\beta) w_h(|\mathbf{x}_\alpha - \mathbf{x}_\beta|) \Big) V_\beta \Big) = \\ &= \sum_{\beta=1}^{N} \mathbf{F}(t, \mathbf{x}_\beta) \cdot \nabla_\alpha w_h(|\mathbf{x}_\alpha - \mathbf{x}_\beta|) V_\beta = \sum_{\beta=1}^{N} \mathbf{F}_\beta \cdot \nabla_\alpha w_{\alpha\beta} V_\beta.\end{aligned}$$

1.3 Classical SPH

One of the pillars of SPH's success lies in its flexibility. Not only can SPH be used to discretise a broad range of PDEs, but the *same* system can often be discretised in several different ways, giving the freedom to choose a problem-specific formulation.[11]

In this and the following sections, we will focus on different ways to discretise the Euler system **??** → p. ??.

Improved discretisation

In the previous, we have derived discrete versions of the grad and div operators

$$\nabla\Phi(t,\mathbf{x}_\alpha) \approx \sum_{\beta=1}^{N} \Phi_\beta \nabla_\alpha w_{\alpha\beta} V_\beta, \tag{1.11}$$

$$\nabla_\alpha \cdot \mathbf{F}(t,\mathbf{x}_\alpha) \approx \sum_{\beta=1}^{N} \mathbf{F}_\beta \cdot \nabla_\alpha w_{\alpha\beta} V_\beta. \tag{1.12}$$

However, this is not a unique way to discretise the differential operators — and particularly *not a good one* either. To demonstrate, take the constant fields $\Phi(t,\mathbf{x}) = 1, \mathbf{F}(t,\mathbf{x}) = \mathbf{e}_x$, then

$$\nabla_\alpha\Phi(t,\mathbf{x}_\alpha) = 0 \neq \sum_{\beta=1}^{N} \Phi_\beta \nabla_\alpha w_{\alpha\beta} V_\beta = \sum_{\beta=1}^{N} \nabla_\alpha w_{\alpha\beta} V_\beta,$$
$$\nabla_\alpha \cdot \mathbf{F}(t,\mathbf{x}_\alpha) = 0 \neq \sum_{\beta=1}^{N} \mathbf{F}_\beta \cdot \nabla_\alpha w_{\alpha\beta} V_\beta = \sum_{\beta=1}^{N} \mathbf{e}_x \cdot \nabla_\alpha w_{\alpha\beta} V_\beta.$$

The naive discretisation Equation (1.11) → p. 7, Equation (1.12) → p. 7 fails to approximate constant fields. To refine the approach, we draw inspiration p. 10 [10] from the identities

$$\nabla\Phi = \phi\nabla\left(\frac{\Phi}{\phi}\right) + \frac{\Phi}{\phi}\nabla\phi, \tag{1.13}$$

$$\nabla\cdot\mathbf{F} = \frac{1}{\phi}\nabla\cdot(\phi\mathbf{F}) - \frac{1}{\phi}\mathbf{F}\cdot\nabla\phi, \tag{1.14}$$

valid for a nonzero continuously differentiable function ϕ.[12] We can improve our approximation by discretising in the following manner

$$\begin{aligned}
\nabla_\alpha\Phi_\alpha &= \phi_\alpha\nabla_\alpha\left(\frac{\Phi_\alpha}{\phi_\alpha}\right) + \frac{\Phi_\alpha}{\phi_\alpha}\nabla_\alpha(\phi_\alpha) \approx \\
&\approx \phi_\alpha \sum_{\beta=1}^{N} \frac{\Phi_\beta}{\phi_\beta}\nabla_\alpha w_{\alpha\beta} V_\beta + \frac{\Phi_\alpha}{\phi_\alpha}\sum_{\beta=1}^{N}\phi_\beta\nabla_\alpha w_{\alpha\beta} V_\beta = \\
&= \phi_\alpha \sum_{\beta=1}^{N} \frac{\Phi_\beta}{\phi_\beta^2}\phi_\beta\nabla_\alpha w_{\alpha\beta} V_\beta + \phi_\alpha\sum_{\beta=1}^{N}\frac{\Phi_\alpha}{\phi_\alpha^2}\phi_\beta\nabla_\alpha w_{\alpha\beta} V_\beta = \\
&= \phi_\alpha \sum_{\beta=1}^{N}\left(\frac{\Phi_\alpha}{\phi_\alpha^2} + \frac{\Phi_\beta}{\phi_\beta^2}\right)\nabla_\alpha w_{\alpha\beta}\phi_\beta V_\beta,
\end{aligned}$$

and

$$
\begin{aligned}
\nabla_\alpha \cdot \mathbf{F}_\alpha &= \frac{1}{\phi_\alpha} \nabla_\alpha \cdot (\phi_\alpha \mathbf{F}_\alpha) - \frac{1}{\phi_\alpha} \mathbf{F}_\alpha \cdot \nabla_\alpha \phi_\alpha \approx \\
&\approx \frac{1}{\phi_\alpha} \sum_{\beta=1}^{N} \phi_\beta \mathbf{F}_\beta \cdot \nabla_\alpha w_{\alpha\beta} V_\beta - \frac{1}{\phi_\alpha} \mathbf{F}_\alpha \cdot \sum_{\beta=1}^{N} \phi_\beta \nabla_\alpha w_{\alpha\beta} V_\beta = \\
&= \frac{1}{\phi_\alpha} \sum_{\beta=1}^{N} (\mathbf{F}_\beta - \mathbf{F}_\alpha) \cdot \nabla_\alpha w_{\alpha\beta} \phi_\beta V_\beta.
\end{aligned}
$$

Thanks to the $\mathbf{F}_\beta - \mathbf{F}_\alpha$ term in the approximate divergence operator, we see that the discretised divergence operator correctly vanishes for constant fields. The alternate discretised gradient operator does *not vanish* for constant fields, however, it possesses a symmetric property in the summands, which will prove useful in what follows.

All in all, the discretisation

$$
\nabla_\alpha \Phi(t, \mathbf{x}_\alpha) \approx \phi_\alpha \sum_{\beta=1}^{N} \left(\frac{\Phi_\alpha}{\phi_\alpha^2} + \frac{\Phi_\beta}{\phi_\beta^2} \right) \phi_\beta \nabla_\alpha w_{\alpha\beta} \phi_\beta V_\beta, \tag{1.15}
$$

$$
\nabla_\alpha \cdot \mathbf{F}(t, \mathbf{x}_\alpha) \approx \frac{1}{\phi_\alpha} \sum_{\beta=1}^{N} (\mathbf{F}_\beta - \mathbf{F}_\alpha) \cdot \nabla_\alpha w_{\alpha\beta} \phi_\beta V_\beta, \tag{1.16}
$$

will be preferred.

Discrete balance of mass

Using the improved divergence operators Equation (1.16) → p. 8 with the auxiliary field ϕ taken as the density ρ the balance of mass

$$
\dot{\rho} = -\rho (\nabla \cdot \mathbf{v}),
$$

can be rewritten as

$$
\frac{\mathrm{d}\rho_\alpha}{\mathrm{d}t} = -\frac{\rho_\alpha}{\rho_\alpha} \sum_{\beta=1}^{N} (\mathbf{v}_\beta - \mathbf{v}_\alpha) \cdot \nabla_\alpha w_{\alpha\beta} \rho_\beta V_\beta = \sum_{\beta=1}^{N} m_\beta (\mathbf{v}_\alpha - \mathbf{v}_\beta) \cdot \nabla_\alpha w_{\alpha\beta},
$$

where we have used a *possible definition* of the particle's density $\rho_\alpha = \frac{m_\alpha}{V_\alpha}$. The discrete balance of mass

$$
\frac{\mathrm{d}\rho_\alpha}{\mathrm{d}t} = \sum_{\beta=1}^{N} m_\beta (\mathbf{v}_\alpha - \mathbf{v}_\beta) \cdot \nabla_\alpha w_{\alpha\beta}, \tag{1.17}
$$

is thus a system of N ordinary differential equations, whose solution yields the densities ρ_α.

This is not the only way we can obtain the densities. An *alternative* formulation defines the density directly as a kernel sum: notice that the properties of the interpolation kernel w_h Table 1.3 →p.2 allow us to interpret the sum

$$\rho_\alpha = \sum_{\beta=1}^{N} m_\beta w_{\alpha\beta}, \tag{1.18}$$

as a *weighted average* of the particle masses.[13] With this approach, we replace the need to numerically solve the ODEs Equation (1.17) →p.9 by a summation procedure.[14] This approach becomes particularly useful in formulations where the density field is not the primary quantity of interest; for diagnostic purposes, one can recover the densities at an arbitrary integration step with a single sum rather than solving for the densities all along. Moreover, [16] has shown that, with a suitable integrator, numerical simulations of isentropic flows in this formulation are *time-reversible*.

Note that the two presented formulations are *not equivalent* — in general, they result in different densities.

Discrete balance of linear momentum

Let us consider the *improved* discrete versions of the gradient Equation (1.15) →p.8 operator and choose again the auxiliary field ϕ as the density ρ. Then the balance of momentum

$$\dot{\mathbf{v}} = -\frac{1}{\rho}\nabla p + \mathbf{g},$$

can be recast to the form

$$\frac{\mathrm{d}\mathbf{v}_\alpha}{\mathrm{d}t} = -\frac{1}{\rho_\alpha}\rho_\alpha \sum_{\beta=1}^{N}\left(\frac{p_\alpha}{\rho_\alpha^2} + \frac{p_\beta}{\rho_\beta}^2\right)\rho_\beta V_\beta \nabla_\alpha w_{\alpha\beta} = \sum_{\beta=1}^{N} m_\beta \left(\frac{p_\alpha}{\rho_\alpha}^2 + \frac{p_\beta}{\rho_\beta}^2\right)\nabla_\alpha w_{\alpha\beta},$$

where we have used the fact $\rho_\alpha = \frac{m_\alpha}{V_\alpha}$. Since $\rho_\alpha = \rho_\alpha(t)$ is a function of time only, the material time derivative in fact becomes an *ordinary* time derivative.[15]

Notice the structure of the discretised balance of linear momentum. If we denote

$$\mathbf{F}_{\beta\to\alpha} \coloneqq -m_\alpha m_\beta \left(\frac{p_\alpha}{\rho_\alpha^2} + \frac{p_\beta}{\rho_\beta^2}\right)\nabla_\alpha w_{\alpha\beta},$$
$$\mathbf{F}_\alpha \coloneqq m_\alpha \mathbf{g},$$

then we can write

$$
\begin{aligned}
m_\alpha \frac{\mathrm{d}\mathbf{v}_\alpha}{\mathrm{d}t} &= -m_\alpha \sum_{\beta=1}^{N} m_\beta \left(\frac{p_\alpha}{\rho_\alpha^2} + \frac{p_\beta}{\rho_\beta^2} \right) \nabla_\alpha w_{\alpha\beta} + \mathbf{g} = \\
&= -\sum_{\beta=1}^{N} m_\alpha m_\beta \left(\frac{p_\alpha}{\rho_\alpha^2} + \frac{p_\beta}{\rho_\beta^2} \right) \nabla_\alpha w_{\alpha\beta} + \mathbf{g} = \sum_{\beta=1}^{N} \mathbf{F}_{\beta\to\alpha} + \mathbf{F}_\alpha .
\end{aligned}
$$

We see that $\mathbf{F}_{\beta\to\alpha}$ is the *internal* pressure force through which the particle β acts on the particle α and $\mathbf{F}_\alpha$ is the gravity force acting on the particle α, i.e., an *external* force acting on the particle α. Moreover, observe that it holds

$$
\mathbf{F}_{\beta\to\alpha} = -m_\alpha m_\beta \left(\frac{p_\alpha}{\rho_\alpha^2} + \frac{p_\beta}{\rho_\beta^2} \right) \nabla_\alpha w_{\alpha\beta} = m_\beta m_\alpha \left(\frac{p_\beta}{\rho_\beta^2} + \frac{p_\alpha}{\rho_\alpha^2} \right) \nabla_\beta w_{\beta\alpha} = -\mathbf{F}_{\alpha\to\beta},
$$

where we have used $w_{\alpha\beta} = w_{\beta\alpha}, \nabla_\alpha w_{\alpha\beta} = -\nabla_\beta w_{\alpha\beta}$, both due to the radial symmetry of w_h.[16] Therefore,

$$
\mathbf{F}_{\beta\to\alpha} = -\mathbf{F}_{\alpha\to\beta},
$$

and we have recovered *Newton's third law* at the *discrete level.* In an isolated system, i.e., in the absence of external forces, Newton's third law implies the *conservation of total linear momentum.*[17] This conservation property would *not hold* had we used the naive discrete gradient Equation (1.11) → p.7 instead of the improved form Equation (1.15) → p.8.

Complete system

To close the discretised equations, we require an equation of state for the pressure $p = p(\rho, T)$. Consistent with our application in atmospheric dynamics, we deploy the thermal equation of a calorically ideal gas ?? → p.??. Also, since we are interested in *isentropic* flows, we add the equation $\dot{\theta}_\alpha = 0$ for the evolution of the potential temperature. Our complete system hence reads as

$$
\frac{\mathrm{d}\mathbf{v}_\alpha}{\mathrm{d}t} = -\sum_{\beta=1}^{N} m_\beta \left(\frac{p_\alpha}{\rho_\alpha^2} + \frac{p_\beta}{\rho_\alpha^2} \right) + \mathbf{g}, \tag{1.19}
$$

$$
p_\alpha(\rho_\alpha, T_\alpha) = R\rho_\alpha T_\alpha, \tag{1.20}
$$

$$
\theta_\alpha = \text{const}, \tag{1.21}
$$

together with an equation for density Equation (1.17) → p.9

$$\frac{\mathrm{d}\rho_\alpha}{\mathrm{d}t} = \sum_{\beta=1}^{N} m_\beta (\mathbf{v}_\alpha - \mathbf{v}_\beta) \cdot \nabla_\alpha w_{\alpha\beta},$$

or Equation (1.18) → p. 9

$$\rho_\alpha = \sum_{\beta=1}^{N} m_\beta w_{\alpha\beta}.$$

In the above equations, the index $\alpha \in \{1, \ldots, N\}$ runs over all particles.

1.4 Hamiltonian formulation

The conservation property we discovered in the previous section aligns with our intuition about the correspondence between SPH and the classical mechanics of point particles. Classical mechanics can be elegantly reframed in the language of *Hamiltonian mechanics*, and we now develop the Hamiltonian formulation of SPH. This reformulation naturally motivates the use of a *symplectic integrator* in the numerical solution of the system of ODEs obtained by the SPH discretisation.

Poisson brackets

Let us begin with a general description of continuum mechanics in the framework of Hamiltonian mechanics. We denote by $\mathbf{X} \in \Omega_0$ the elements of the Lagrangian, i.e., reference, configuration manifold Ω_0 and by $\mathbf{x}$ the elements of the Eulerian, i.e., current, configuration manifold. As usual, the deformation (motion) of the continuum is given as a C^1 diffeomorphism $\boldsymbol{\chi} : (t, \mathbf{X}) \mapsto \mathbf{x}$, i.e., $\mathbf{x} = \boldsymbol{\chi}(t, \mathbf{X})$ from the Lagrangian to the Eulerian configuration. Denote further $\mathbf{M}(t, \mathbf{X})$ the Eulerian momentum density per Lagrangian volume.. The mappings $\mathbf{x}(t, \mathbf{X}), \mathbf{M}(t, \mathbf{X})$ will be called the *Lagrangian state variables*.

The continuum counterpart to the canonical Poisson bracket of Hamiltonian mechanics of points is the *canonical* Poisson bracket in the Lagrangian frame; given two arbitrary Fréchet differentiable functionals $F(t, \mathbf{x}(t, \mathbf{X}), \mathbf{M}(t, \mathbf{X}))$, $G(t, \mathbf{x}(t, \mathbf{X}), \mathbf{M}(t, \mathbf{X}))$ of the Lagrangian state variables $\mathbf{x}, \mathbf{M}$, we define p. 2 [17]

$$\{F, G\}^{(\mathrm{L})} := \int_{\Omega_0} \left(\frac{\delta F}{\delta x^i(\mathbf{X})} \frac{\delta G}{\delta M_i(\mathbf{X})} - \frac{\delta F}{\delta M_i(\mathbf{X})} \frac{\delta G}{\delta x^i}(\mathbf{X}) \right) \mathrm{d}\mathbf{X}\,. \tag{1.22}$$

The derivatives appearing in Equation (1.22) → p. 12 are functional derivatives, i.e., distributions representing the Fréchet differential. For a concise introduction, see e.g., the appendix of [18].

In what follows, we will for brevity of notation omit the dependence of $\mathbf{x}(t, \mathbf{X}), \mathbf{M}(t, \mathbf{X})$ on $(t, \mathbf{X})$ and simply write $\mathbf{x}, \mathbf{M}$. One can check that the mapping $(F, G) \mapsto \{F, G\}^{(\mathrm{L})}$ is in fact a skew-symmetric bilinear form that satisfies the Jacobi identity and Leibniz rule.[18]

Given an Fréchet differentiable functional $F = F(t, \mathbf{x}, \mathbf{M})$, its evolution in time is given by p. 2 [17]

$$\dot{F} = \partial_t F + \{F, E^{(\mathrm{L})}\}^{(\mathrm{L})}, \tag{1.23}$$

where $E^{(\mathrm{L})}$ is the functional of total energy expressed using the Lagrangian state variables $\mathbf{x}, \mathbf{M}$.[19] In particular, for a functional $F(\mathbf{x}, \mathbf{M})$ that explicitly does not depend on time, we can calculate using the chain rule

$$\dot{F}(\mathbf{x}, \mathbf{M}) = \int_{\Omega_0} \left(\frac{\delta F}{\delta x^i} \, \partial_t x^i + \frac{\delta F}{\delta M_i} \, \partial_t M_i \right) \mathrm{d}\mathbf{X} \,,$$

on the other hand Equation (1.23) → p. 12 dictates

$$\dot{F}(\mathbf{x}, \mathbf{M}) = \int_{\Omega_0} \left(\frac{\delta F}{\delta x^i} \, \frac{\delta E^{(\mathrm{L})}}{\delta M_i} - \frac{\delta F}{\delta M_i} \, \frac{\delta E^{(\mathrm{L})}}{\delta x^i} \right) \mathrm{d}\mathbf{X} \,,$$

since F were arbitrary, we conclude p. 3 [17] that it must hold

$$\partial_t x^i(t, \mathbf{X}) = \frac{\delta E^{(\mathrm{L})}}{\delta M_i(t, \mathbf{X})} \,, \tag{1.24}$$

$$\partial_t M_i(t, \mathbf{X}) = - \frac{\delta E^{(\mathrm{L})}}{\delta x^i(t, \mathbf{X})} \,. \tag{1.25}$$

The equations Equation (1.24) → p. 13, Equation (1.25) → p. 13 are the Hamilton *canonical* equations of continuum mechanics. The right-hand sides of the equations require the knowledge of the total energy of the continuum expressed using the Lagrangian state variables.

In general, the energy has a contribution from the kinetic energy $|\mathbf{M}|^2/2\rho_0$ and the internal energy e. The fundamental thermodynamic relation $e = e(\rho, \eta)$, η being the entropy density is naturally expressed in terms of *Eulerian* volume densities ρ and η. To incorporate it into the Lagrangian energy functional, we transform to the Lagrangian frame using the Jacobian $J(\mathbf{X}) = \det \nabla \boldsymbol{\chi}(\mathbf{X})$

$$E^{(\mathrm{L})} = \int_{\Omega_0} \left(\frac{|\mathbf{M}|^2}{2\rho_0} + Je(\rho(\mathbf{x}(\mathbf{X}))), \eta(\mathbf{x}(\mathbf{X})) \right) \mathrm{d}\mathbf{X} \,. \tag{1.26}$$

We briefly summarize the structure of what follows. We have noticed that for the description of the evolution of a functional of the Lagrangian state variables, we require **(1)** the Poisson bracket Equation (1.22) → p.12, **(2)** the total energy functional Equation (1.26) → p.13. To obtain the time evolution of a functional of the *SPH state variables*, we require **(1)** a SPH Poisson bracket, **(2)** the SPH energy . The SPH bracket and energy are obtained by reduction and approximation of their continuum counterparts respectively.

SPH discretisation

Denote $\{\Omega_\alpha\}_{\alpha=1}^N, N \in \mathbb{N}$ a pairwise disjoint finite partition of the reference manifold Ω_0, i.e., $\Omega_0 = \bigcup_{\alpha=1}^N \Omega_\beta$, $\Omega_\alpha \cap \Omega_\beta = \emptyset$ for $\alpha \neq \beta$. The sets Ω_α will represent the *SPH particles*. Moreover, with the aid of the normalized characteristic function of Ω_α,

$$\chi_\alpha(\mathbf{X}) = \begin{cases} \dfrac{1}{V_\alpha}, & \mathbf{X} \in \Omega_\alpha, \\ 0, & \mathbf{X} \notin \Omega_\alpha \end{cases}, V_\alpha = \int_{\Omega_\alpha} \mathrm{d}\mathbf{X} \,,$$

we can define the *SPH position* $\mathbf{x}_\alpha(t)$ as

$$\mathbf{x}_\alpha(t) = \int_{\Omega_0} \mathbf{x}(t, \mathbf{X}) \chi_\alpha(\mathbf{X}) \, \mathrm{d}\mathbf{X} \,, \tag{1.27}$$

and the *SPH momentum* $\mathbf{M}_\alpha(t, \mathbf{X})$ as

$$\mathbf{M}_\alpha(t) = \int_{\Omega_0} \mathbf{M}(t, \mathbf{X}) \chi_\alpha(\mathbf{X}) \, \mathrm{d}\mathbf{X} \,. \tag{1.28}$$

Naturally, the SPH positions and momenta $(\mathbf{x}_\alpha, \mathbf{M}_\alpha)$ will replace the state variables $(\mathbf{x}, \mathbf{M})$ of the continuum mechanics, and so we will refer to them as the SPH state variables.

SPH bracket

Let $F(\mathbf{x}_\alpha, \mathbf{M}_\alpha), G(\mathbf{x}_\alpha, \mathbf{M}_\alpha)$ be two arbitrary Fréchet differentiable functionals of the SPH state variables. From the definition of the SPH state variables Equation (1.27) → p.13, Equation (1.28) → p.14 it follows that F, G in fact also depend on the state variables of the continuum mechanics —hence we are allowed to plug

them into the Lagrangian Poisson bracket Equation (1.22) → p. 12. The calculation reads as

$$\{F,G\}^{(\mathrm{L})} = \int_{\Omega_0} \left(\frac{\delta F}{\delta x^i} \frac{\delta G}{\delta M_i} - \frac{\delta F}{\delta M_i} \frac{\delta G}{\delta x^i} \right) \mathrm{d}\mathbf{X} =$$

$$= \sum_{\alpha,\beta} \int_{\Omega_0} \left(\left(\frac{\partial F}{\partial x^j_\alpha} \frac{\delta x^j_\alpha}{\delta x^i} + \frac{\partial F}{\partial M_{\beta,j}} \frac{\delta M_{\beta,j}}{\delta x^i} \right) \left(\frac{\partial G}{\partial M_{\beta,j}} \frac{\delta M_{\beta,j}}{\delta M_i} + \frac{\partial G}{\partial x^j_\alpha} \frac{\delta x^j_\alpha}{\delta M_i} \right) \right) \mathrm{d}\mathbf{X} +$$

$$- \sum_{\alpha,\beta} \int_{\Omega_0} \left(\left(\frac{\partial F}{\partial M_{\beta,j}} \frac{\delta M_{\beta,j}}{\delta M_i} + \frac{\partial F}{\partial x^j_\alpha} \frac{\delta x^j_\alpha}{\delta M_i} \right) \left(\frac{\partial G}{\partial x^j_\alpha} \frac{\delta x^j_\alpha}{\delta x^i} + \frac{\partial G}{\partial M_{\beta,j}} \frac{\delta M_{\beta,j}}{\delta x_i} \right) \right) \mathrm{d}\mathbf{X} \,.$$

Our goal is to simplify **??** → p. ?? and obtain an expression depending only on the SPH state variables $\mathbf{x}_\alpha, \mathbf{M}_\alpha$. The procedure requires the evaluation of the functional derivatives of x^j_α, $M_{\beta,j}$ w.r.t M_i, x^i. From the definition Equation (1.28) → p. 14, we obtain for any $\varphi \in C_0^\infty(\Omega_0)$

$$\left\langle \frac{\delta x^j_\alpha}{\delta x^i(\mathbf{X})}, \varphi \right\rangle = \int_{\Omega_0} \frac{\delta x^j_\alpha}{\delta x^i} \varphi \, \mathrm{d}\mathbf{X} = \frac{\mathrm{d}}{\mathrm{d}s} x^j_\alpha(x^i + s\varphi)|_{s=0} =$$

$$= \frac{\mathrm{d}}{\mathrm{d}s} \int_{\Omega_0} \delta^j_i (x^i + s\varphi) \chi_\alpha \, \mathrm{d}\mathbf{X}|_{s=0} = \int_{\Omega_0} \delta^j_i \chi_\alpha \varphi \, \mathrm{d}\mathbf{X} \,,$$

and so

$$\frac{\delta x^j_\alpha}{\delta x^i} = \delta^j_i \chi_\alpha .$$

In the same manner we obtain

$$\frac{\delta M_{\beta,j}}{\delta M_i} = \delta^j_i \chi_\beta ,$$

and the cross terms $\frac{\delta x^j_\alpha}{\delta M_i}, \frac{\delta M_{\beta,j}}{\delta x^i}$ are zero, as the definition of the SPH position Equation (1.27) → p. 13 and SPH momentum Equation (1.28) → p. 14 include only the continuous position $\mathbf{x}$ and continuous momentum $\mathbf{M}$ respectively.

The expression **??** → p. ?? can be rewritten to obtain

$$\{F(\mathbf{x}_\alpha, \mathbf{M}_\alpha), G(\mathbf{x}_\alpha, \mathbf{M}_\alpha)\}^{(\mathrm{L})} = \sum_\alpha \left(\frac{\partial F}{\partial x^i_\alpha} \frac{\partial G}{\partial M_{\alpha,i}} - \frac{\partial F}{\partial M_{\alpha,i}} \frac{\partial G}{\partial x^i_\alpha} \right).$$

It is hence natural to define the *SPH bracket* for any two functionals $F(\mathbf{x}_\alpha, \mathbf{M}_\alpha), G(\mathbf{x}_\alpha, \mathbf{M}_\alpha)$ of the SPH state variables through.

$$\{F(\mathbf{x}_\alpha, \mathbf{M}_\alpha), G(\mathbf{x}_\alpha, \mathbf{M}_\alpha)\}^{(\mathrm{SPH})} := \sum_\alpha \left(\frac{\partial F}{\partial x_\alpha^i} \frac{\partial G}{\partial M_{\alpha,i}} - \frac{\partial F}{\partial M_{\alpha,i}} \frac{\partial G}{\partial x_\alpha^i} \right), \tag{1.29}$$

SPH energy functional

Now we aim to rewrite Equation (1.26) → p. 13 in terms of the SPH state variables. We begin with the definition of the Lagrangian density of the particle

$$\rho_\alpha^{(\mathrm{L})} = \frac{m_\alpha}{V_\alpha}, \tag{1.30}$$

where

$$m_\alpha = \int_\Omega \rho^{(\mathrm{L})}(\mathbf{X}) \chi_\alpha(\mathbf{X}) \, \mathrm{d}\mathbf{X}. \tag{1.31}$$

Assume the density can be expressed[20] as a function of density $\rho^{(\mathrm{L})}$, and entropy $\eta^{(\mathrm{L})}$. $e^{(\mathrm{L})} = e^{(\mathrm{L})}(\rho^{(\mathrm{L})}, \eta^{(\mathrm{L})})$. Using the transformation of density and the extensively of the internal energy, we see that

$$e^{(\mathrm{L})}(\mathbf{X}) = e(\rho^{(\mathrm{L})}(\mathbf{X})) = e(J(\mathbf{X})\rho(\mathbf{x}(\mathbf{X}))) = J(\mathbf{X}) e(\rho(\mathbf{x}(\mathbf{X}))),$$

where $J = \det \nabla \boldsymbol{\chi}$ and the notational shortcut $\mathbf{x}(\mathbf{X})$ as usual means $\mathbf{x}(\mathbf{X}) = \boldsymbol{\chi}(t, \mathbf{X})$.[21]

Classical SPH as a Hamiltonian system

Symplectic integrators

1.5 Density-free formulations

Pressure-entropy

Recall that the entropy of an isentropic flow of an ideal gas has the form[22]

$$\eta = \eta_0 + c_v \log\left(\frac{P}{\rho^\gamma}\right)$$

and $\eta =$ const. We see that the quantity A defined as [19]

$$A = \frac{P}{\rho^\gamma}\,,$$

is in fact equal to

$$A = \exp\left(\frac{\eta - \eta_0}{c_v}\right),$$

meaning that the flow is *isentropic*, if and only if the quantity A is *conserved* under the flow. Thus A is an *entropy-like* variable: conserved under isentropic flows.

The choice

$$x_\alpha = m_\alpha A_\alpha^{\frac{1}{\gamma}}, \tag{1.32}$$

gives the pressure P_α as

$$P_\alpha = \left(\sum_{\beta=1}^{N} m_\beta A_\beta^{\frac{1}{\gamma}} w_{\alpha\beta}(h_\alpha)\right)^\gamma \tag{1.33}$$

and the evolution equations as

$$\frac{\mathrm{d}\mathbf{v}_\alpha}{\mathrm{d}t} = -\sum_{\beta=1}^{N} m_\beta (A_\alpha A_\beta)^{\frac{1}{\gamma}} \left(\frac{f_\alpha P_\alpha}{P_\alpha^{\frac{2}{\gamma}}} \nabla_\alpha w_{\alpha\beta}(h_\alpha) + \frac{f_\beta P_\beta}{P_\beta^{\frac{2}{\gamma}}} \nabla_\alpha w_{\alpha\beta}(h_\beta)\right), \tag{1.34}$$

with

$$f_\alpha = \left(1 + \frac{h_\alpha}{2n_\alpha}\frac{\partial n_\alpha}{\partial h_\alpha}\right)^{-1}$$

Pressure-potential temperature

Recall the *potential temperature* θ

$$\theta = T\left(\frac{P_0}{P}\right)^{\frac{R}{c_p}},$$

where T is the gas's thermodynamic temperature, and P_0 is the reference pressure. [23] Using the thermal state equation

$$P = R\rho T,$$

together with the fact

$$\frac{R}{c_p} = \frac{\gamma - 1}{\gamma},$$

we obtain

$$\theta = \frac{P}{\rho R} P_0^{\frac{\gamma-1}{\gamma}} P^{\frac{1-\gamma}{\gamma}} = \frac{P_0^{\frac{\gamma-1}{\gamma}}}{R} \left(\frac{P^{\frac{1}{\gamma}}}{\rho} \right) = \frac{P_0^{\frac{\gamma-1}{\gamma}}}{R} \left(\frac{P}{\rho^\gamma} \right)^{\frac{1}{\gamma}} = \frac{P_0^{\frac{\gamma-1}{\gamma}}}{R} A^{\frac{1}{\gamma}},$$

so setting

$$\tilde{\theta}_0 := \frac{P_0^{\frac{\gamma-1}{\gamma}}}{R}$$

we obtain

$$\theta = \tilde{\theta}_0 A^{\frac{1}{\gamma}}. \tag{1.35}$$

This implies the potential temperature θ is also *an entropy-like* variable - it is *conserved* under isentropic flows.

Rewriting 1.32 using 1.35 we obtain

$$x_\alpha = \frac{m_\alpha}{\tilde{\theta}_0} \theta_\alpha. \tag{1.36}$$

Using this quantity the pressure 1.33 can then be expressed as

$$P_\alpha = \left(\sum_{\beta=1}^{N} \frac{m_\beta}{\tilde{\theta}_0} \theta_\beta w_{\alpha\beta}(h_\alpha) \right)^\gamma. \tag{1.37}$$

From 1.34 we obtain

$$\frac{\mathrm{d}\mathbf{v}_\alpha}{\mathrm{d}t} = -\sum_{\beta=1}^{N} m_\beta \frac{\theta_\alpha \theta_\beta}{\tilde{\theta}_0^2} \left(\frac{f_\alpha P_\alpha}{P_\alpha^{\frac{2}{\gamma}}} \nabla_\alpha w_{\alpha\beta}(h_\alpha) + \frac{f_\beta P_\beta}{P_\beta^{\frac{2}{\gamma}}} \nabla_\alpha w_{\alpha\beta}(h_\beta) \right), \tag{1.38}$$

with **??** unchanged

$$f_\alpha = \left(1 + \frac{h_\alpha}{2n_\alpha} \frac{\partial n_\alpha}{\partial h_\alpha} \right)^{-1}. \tag{1.39}$$

1.6 Implementation specifics

Extrapolating boundary conditions

Inflow & Outflow

Free-slip

Adaptive smoothing length

Newton iterations

Particle placement

WitchExpGrid